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Course/Program:- S.Y. B.Tech C.S.E

Subject:- Computer organization and Architecture.

Theory assignment - 03.

Q.1) What is Register Transfer Language?

Solⁿ:-

Register Transfer Language (RTL) is a low-level language that is used to describe the functioning of a digital circuit and, more specifically, the transfer of information between registers. It provides how data moves from one register to the other and how data is processed within the digital system. Through RTL, there is a capability of creating abstraction levels where high-level design descriptions can be created and easily linked to low-level hardware implementation in designing, simulating, as well as synthesizing digital circuits.

Key concepts of Register Transfer Language (RTL)

- The meaningful descriptions are provided by RTL for the flow of data between the registers of the hardware.
- It is applied to the representation of Synchronous circuits: circuits that are controlled by clock signals.
- Transfer of data and logical operations are described for the register level.
- As mentioned earlier, RTL designs are documented most often in hardware description languages, including Verilog or VHDL.

- It is an abstraction layer between high-order Applications Specific Integrated Circuits (ASICs) and physical devices.

Usage of RTL in Digital Design.

- Proclaims the nature of hardware at the register-transfer.
- Formerly utilized to model data flow while using registers.
- Assists in generation to gate level of designs.
- It allows simulation and validation of the relevant behaviours of the hardware under design.
- Serves as the foundation for developing circuits within the digital domain with the use of HDLs.

Q.2) Write down the Advantages and disadvantages of Register Transfer Language (RTL)?

Solⁿ:-

→ Advantages of Register Transfer Language (RTL)

- Enables efficient hardware design.
- This make it possible to simulate some activities and perhaps detect some errors at an early date.
- Implements conceptual description up to the gate-level hardware.
- It helps to reuse the design components.
- It gives a clear guide on how to do timing analysis on a given design.

→ Disadvantages of Register Transfer Language (RTL)

- Although the performance is high, this type can be quite challenging to debug.
- These may lead to inefficient large constructions if not well optimized for beneficial use.



- The reasoning behind it is rather hardware-oriented and may cause problems that are hard to comprehend without knowledge of message flow.
- Synthetic results appear to depend on the capabilities of specific tools.
- Compared to high-level descriptions, the amount of abstraction retrieved by this system is very low.

Q.3) Difference between Register and memory?

Sol:-

Register	Memory
1. Registers hold the operands or instructions that CPU is currently processing.	Memory holds the instructions and the data that the currently executing program in CPU requires.
2. Register holds the small amount of data around 32-bit to 64-bits.	Memory of the computer can range from some GB to TB.
3. CPU can operate on register contents at the rate of more than one operations in one clock cycle.	CPU accesses memory at the slower rate than register.
4. Types are Accumulator register, Program counter, Instruction register, Address register etc.	Type of memory are RAM, etc.
5. Registers can be controlled i.e. you can store and retrieve information from them.	Memory is almost not controllable.
6. Registers are faster than memory.	RAM is much slower than registers.



Q.4) Short note on- a) Register. b) Memory? Also write their advantages and disadvantages?

Sol:-

Register..

Registers are the smallest data-holding elements that are built into the processor itself. These are the memory locations that are directly by the processor. It may hold an instruction, a storage address, or any data such as a bit sequence or individual characters. For example, an instruction may specify that the contents of two defined registers be multiplied together and then placed in a specific register.

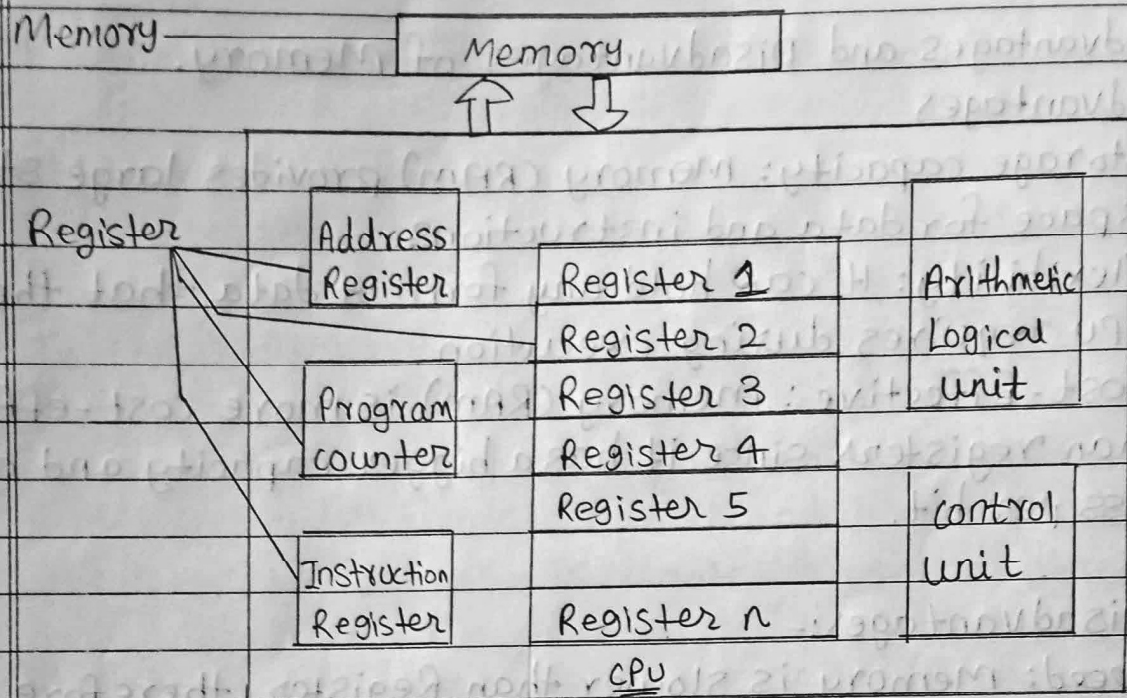
Advantages and Disadvantages of Registers

Advantages:-

- Speed: Registers are the fastest type of storage, providing instant access to data.
- Efficiency:- They allow the CPU to execute instructions faster by storing immediate data.
- Proximity:- Since they are placed within the CPU, they require less time to get data.

Disadvantages:-

- Storage capacity:- Registers have limited storage capacity.
- Manufacture cost:- They are costly to manufacture due to their speed and integration into the CPU.
- Volatility:- Data in registers is lost when the machine is turned off.



Memory

Memory is a hardware device used to store computer programs, instructions and data. The memory that is internal to the processor is a primary memory (RAM), and the memory that is external to the processor is a Secondary memory (Hard Drive).

Memory can also be categorized on the basis of volatile and non-volatile memory. Volatile memory is memory that loses its contents when the computer or hardware devices loses power. RAM (Random Access Memory) is an example of volatile memory.

Non-volatile memory is the memory that keeps its contents even if power gets lost. EPROM is an example of non-volatile memory.

Example: RAM, EPROM etc.



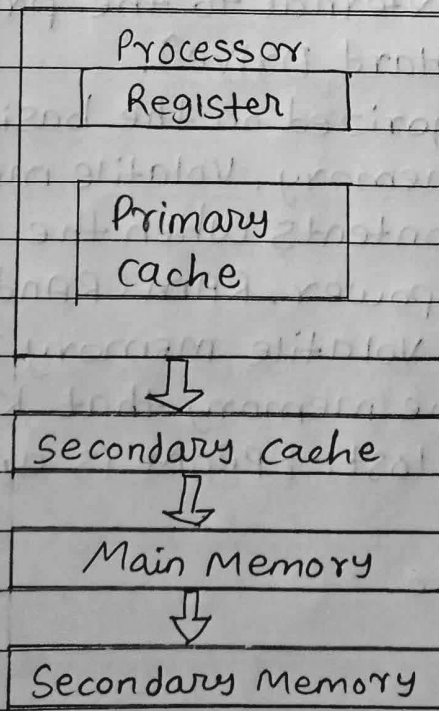
Advantages and Disadvantages of Memory.

Advantages

- **Storage capacity:** Memory (RAM) provides large storage space for data and instructions.
- **Flexibility:** It can hold any form of data that the CPU requires during execution.
- **Cost-Effective:** Memory (RAM) is more cost-effective than registers since it has a bigger capacity and costs less per bit.

Disadvantages:-

- **Speed:** Memory is slower than Register, therefore data retrieval takes longer.
- **Volatility:** -RAM, like registers, is volatile, meaning it loses data when the machine is turned off.
- **Latency:** - Accessing data from memory takes longer, potentially slowing down operations.





Q.5) Explain Shift Micro-operations in detail?

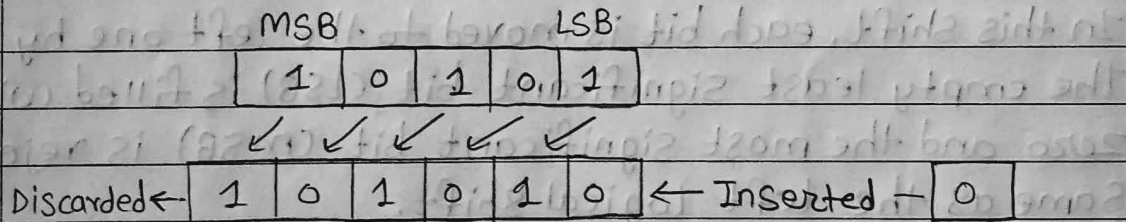
Solⁿ:-

Shift Micro-operations.

(i) Logical shift:- It transfers the 0 zero the serial input.

(ii) Logical left shift:-

In this shift, one position moves each bit to the left one by one. The empty least significant bit (LSB) is filled with zero, and the most significant bit (MSB) is rejected. The left shift operator is denoted by the double left arrow key ($\ll$). The general syntax for the left shift is shift-expression $\ll K$.



Logical left shift

(iii) Logical right shift:-

In this shift, each bit moves to the right one by one and the least significant bit (LSB) is rejected and the empty MSB is filled with zero.

The right shift operator is denoted by the double right arrow key ($\gg$). The general syntax for the right shift is "shift-expression $\gg K$ ".



MSB				LSB	
1	0	1	0	1	

0 - inserted → 0 1 0 1 0 1 → Discarded

Logical right shift.

② Arithmetic Shift.

The arithmetic shift micro-operation moves the signed binary number either to the left or to the right position.

(i) Arithmetic left shift.

In this shift, each bit is moved to the left one by one. The empty least significant bit (LSB) is filled with zero and the most significant bit (MSB) is rejected. Same as the left logical shift.

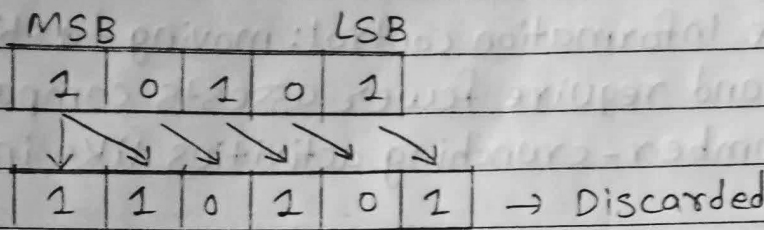
Shift left Arithmetic

7	6	5	4	3	2	1	0
1	0	1	1	0	0	1	1

0	1	1	0	0	1	1	0
							← 1

(ii) Arithmetic right Shift.

In this shift, each bit is moved to the right one by one and the least significant (LSB) bit is rejected and the empty most significant bit (MSB) is filled with the value of the previous MSB.



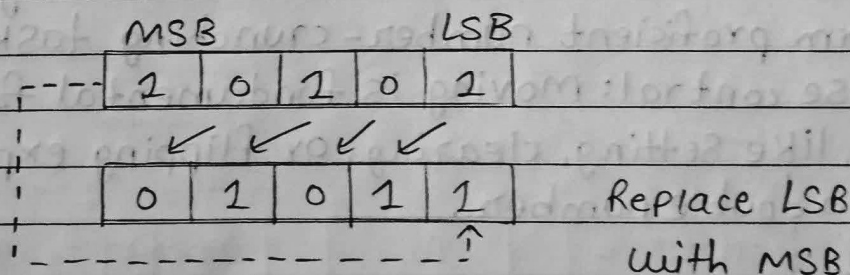
Arithmetic Right Shift.

③ Circular Shift:

The circular shift circulates the bits in the sequence of the register around both ends without any loss of information.

(i) Circular left shift.

In this micro shift operation each bit in the register is shifted to the left one by one. After shifting, the LSB becomes empty, so the value of the MSB is filled in there.



Circular left shift

(ii) Circular right shift:

In this micro shift operation each bit in the register is shifted to the right one by one. After shifting, the MSB becomes empty, so the value of the LSB is filled in there.



Advantages:

- Productive Information control: moving task are frequently quicker and require fewer assests compared with other number-crunching activities like increase or division.
- Bitwise activities: moving tasks are major for bitwise control, which is fundamental for errands like encoding, encryption, and pressure.

Disadvantages:

- Restricted Activity Extension: Shift tasks are just reasonable for explicit sorts of errands.
- Potential for Information Misfortune: Moving pieces can prompt information misfortune in the event that they are not dealt with accurately.

Application:

- Math Activities: Movements are ordinarily used to perform proficient number-crunching tasks.
- Bitwise control: Moving is fundamental for bitwise tasks, like setting, clearing, or flipping explicit pieces inside a double number.

Q.6) Explain an Arithmetic Logic shift in detail?

Solⁿ:-

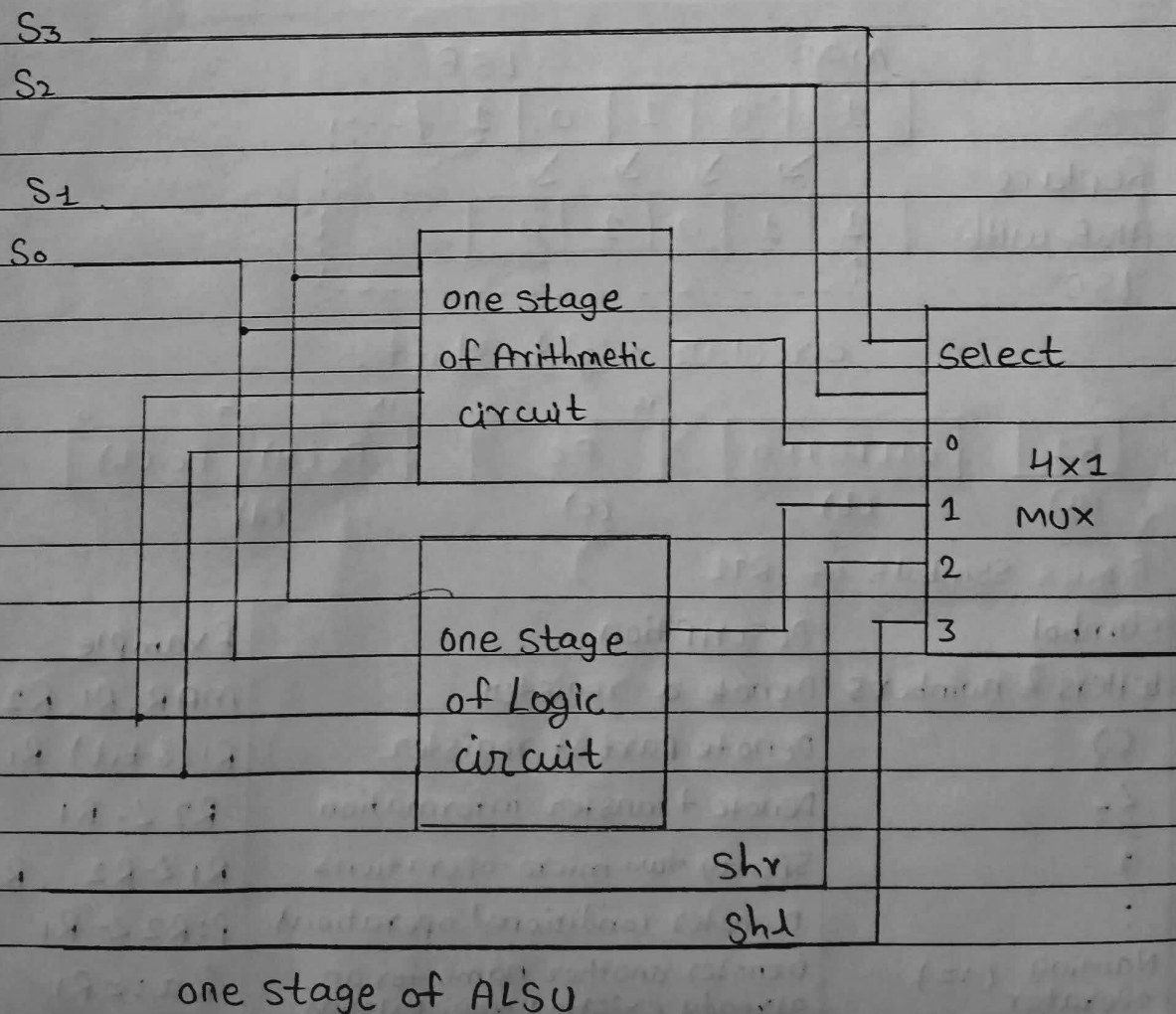
Arithmetic Logic Shift Unit (ALSh) is a member of the Arithmetic Logic unit (ALU) in a computer system. It is a digital circuit that performs logical, arithmetic, and shift operations. Rather than having individual registers calculating the micro operations directly, the computer deploys a number of storage registers which is



connected to a common operational unit known as an arithmetic logic unit or ALU.

Now, to implement the micro operation, the contents of specified registers are allocated in the inputs of the common Arithmetic Logic unit. The Arithmetic Logic Unit performs an operation that leads as a result and gets transferred to a destination register.

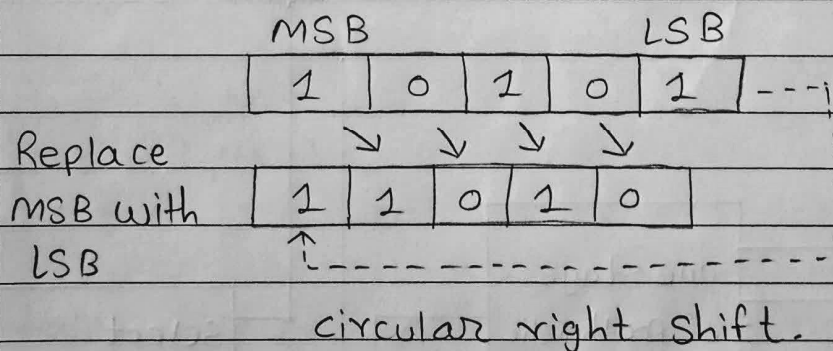
Arithmetic logic unit may be a combinatory circuit in order that the complete register transfer operation from the supply registers through the ALU and into the destination registers is performed throughout one clock pulse amount. Sometimes, the shift micro operations are performed in a separate unit, but sometimes it is made as a part of full ALU.



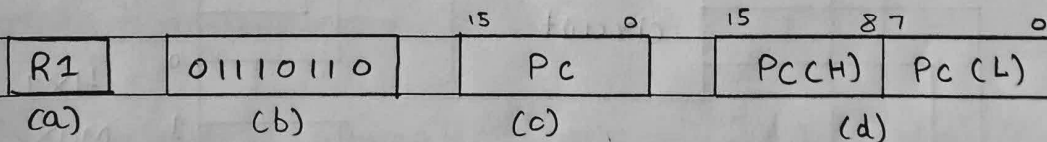


We can combine and make one ALU with common Selection Variables by adding arithmetic, logic, and shifted circuits. We can see the one stage of an arithmetic logic shift unit in the diagram above. Some particular micro operations are selected through the inputs S_1 and S_0 .

4×1 multiplexar at the output chooses between associated arithmetic output between F_i and a logic output in H_i . The data in the multiplexar are selected through inputs S_2 and S_3 and the other two data inputs to the multiplexar obtain the inputs A_{i-1} for the shr operation and A_{i+1} for the shl operations.



Q.1)
contⁿ:



Basics Symbols of RTL

Symbol	Description	Example
letters & numbers	Denote a register	MAR, R_1 , R_2
()	Denote part of register	$R_1(8\text{-bit})$ $R_1(0-7)$
$\leftarrow$	Denote transfer information	$R_2 \leftarrow R_1$
,	Specify two micro-operations	$R_1 \leftarrow R_2, R_2 \leftarrow R_1$
:	Denotes conditional operations	$P: R_2 \leftarrow R_1$ if $P=1$
Naming operator ($:=$)	Denotes another name for an already existing register alias	$Ra := R_1$